

# ZHANG YUTAO

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## Research & Technical Summary

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PhD candidate in Computer Graphics at Université de Montréal, specializing in geometry processing, neural representations, and variational optimization for 3D geometric tasks. Experienced integrating classical numerical methods (Poisson, FEM, Laplacian) with machine learning frameworks for surface reconstruction. Work accepted at SIGGRAPH Asia 2025.

## Education

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| <b>PhD in Computer Graphics</b><br><i>Université de Montréal</i>    | 2023–Present<br>GPA: <b>4.15/4.30</b> |
| <b>BSc in Computer Science</b><br><i>Université de Montréal</i>     | 2020–2022<br>GPA: <b>4.24/4.30</b>    |
| Excellence Scholarships: UdeM (2021, 2022), American Express (2021) |                                       |
| <b>BA in French</b><br><i>Peking University</i>                     | 2016–2020<br>GPA: <b>3.56/4.00</b>    |

## Research Experience

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**Variational Neural Surfacing of 3D Sketches** SIGGRAPH Asia 2025 (Accepted)

- Reformulated and reparameterized the flow-alignment energy for implicit surface representations.
- Developed a gradient-domain optimization method for flow-aligned surfacing from 3D sketches and clean curve networks.

**Keywords:** Neural Implicit Surfacing, Variational Optimization, 3D Sketch Processing.

**Surfacing 3D Sketched Curve Networks with Minimal Currents** 2022–2024

- Investigated minimal surface reconstruction for incomplete or noisy curve networks.
- Applied geodesic measure theory with spatial metric control.

**Keywords:** Minimal Surfaces, Curve Networks, Differential Geometry.

## Expertise Breadth

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**Geometry Processing:** Surface reconstruction, mesh analysis, subdivision, deformation.

**Machine Learning:** Representative learning, reinforcement learning, neural geometric processing, natural language processing, probabilistic graphical models.

**Rendering & Image Synthesis:** Ray tracing, shading, texture filtering, anti-aliasing.

## Course Projects

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**Screened Poisson Surface Reconstruction (Python Implementation)**

- Built a simplified screened Poisson reconstruction for uniform point samples.
- Added smoothing regularization for stable surface recovery.

**Second-Order EWA Texture Filtering**

- Improved elliptical weighted average texture mapping.
- Corrected sampling center after perspective projection using second-order derivatives.

## Technical Skills

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**Programming:** Python (proficient); C++, Java, JavaScript, HTML (familiar); Haskell, Prolog (exposure).  
**Machine Learning:** PyTorch (familiar); JAX (exposure).  
**Graphics & Geometry:** Houdini (familiar); WebGL, Blender (exposure).  
**Mathematics:** Differential geometry; optimization; probability theory.

## Publications

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Zhang, Y., Wang, S., Bessmeltsev, M. *Variational Neural Surfacing of 3D Sketches*. SIGGRAPH Asia 2025 (Accepted).

## Industry & Internship Experience

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| <b>Sinopec Engineering Group (SEG) — Information Center Intern</b>                                                                                                                                 | 2019 |
| <ul style="list-style-type: none"><li>• Learned enterprise data workflows using SAP BW and Oracle databases.</li><li>• Developed and maintained an internal corporate reporting website.</li></ul> |      |
| <b>Foreign Language Teaching &amp; Research Press — Editor/Proofreader</b>                                                                                                                         | 2018 |
| <ul style="list-style-type: none"><li>• Edited and proofread French dictionary software content (text and audio).</li></ul>                                                                        |      |

## Languages

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Chinese (Native) | English (TOEFL 103) | French (B2) | Japanese (nearly N3)

## Extracurricular

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Peking University, School of Foreign Languages

- Basketball Team Member.
- Student Union Activities (Public Relations and Mini-program Development).

## Interests

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**Video games:** 20,000+ hours of gaming experience.  
**Music:** Singing; violin.  
**Sports:** Basketball; billiards; tennis; skiing.